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A Relational Theory of Data Governance

ABSTRACT. Data-governance law—the legal regime that regulates how data about people is collected, processed, and used—is the subject of lively theorizing and several proposed legislative reforms. Different theories advance different legal interests in information. Some seek to reassert individual control for data subjects over the terms of their datafication, while others aim to maximize data-subject financial gain. But these proposals share a common conceptual flaw. Put simply, they miss the point of data production in a digital economy: to put people into population-based relations with one another. This relational aspect of data production drives much of the social value and harm of data collection and use in a digital economy.

This Feature advances a theoretical account of data as social relations, constituted by both legal and technical systems. It shows how data relations result in supraindividual legal interests. Properly representing and adjudicating among those interests necessitates far more public and collective (i.e., democratic) forms of governing data production. Individualist data-subject rights cannot represent, let alone address, these population-level effects.

This account offers two insights for data-governance law. First, it better reflects how and why data collection and use produce economic value as well as social harm in the digital economy. This brings the law governing data flows into line with the economic realities of how data production operates as a key input to the information economy. Second, this account offers an alternative normative argument for what makes datafication—the transformation of information about people into a commodity—wrongful. What makes datafication wrong is not (only) that it erodes the capacity for subject self-formation, but instead that it materializes unjust social relations: data relations that enact or amplify social inequality. This account indexes many of the most pressing forms of social informational harm that animate criticism of data extraction but fall outside typical accounts of informational harm. This account also offers a positive theory for socially beneficial data production. Addressing the inegalitarian harms of datafication—and developing socially beneficial alternatives—will require democratizing data social relations: moving from individual data-subject rights to more democratic institutions of data governance.

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These accounts also guide views on how data governance should be reformed. As I discuss in Part III, concerns over autonomy and dignity guide dignitarian efforts to reduce datafication's commodification of inner life and to increase the regulatory oversight of data flows that act back on users in ways that wrongfully undermine their self-will. Concern over loss of control and lack of clear legal rights to data's value motivate propertarian efforts to formalize these rights for data subjects. These reforms and others focus on increasing data subjects' capacity to determine how (and under what conditions) their data is collected, processed, and used.

II. DATA RELATIONS AND THEIR SOCIAL EFFECTS

One way to evaluate different theories of data governance is to examine how such theories conceive of (and propose to act upon) the social-relations structured by data flows.[85] To understand the significance of data's relationality, let us consider with greater specificity how data relates people to one another, how such relations may produce social effects, and which of these relations are (and are not) accorded legal relevance by current and proposed forms of data-governance law.

A. Data Governance's Sociality Problem

In July 2018, privacy activists reported that the Federal Bureau of Investigation (FBI), along with the National Institute of Standards and Technology, were evaluating the effectiveness of tattoo-recognition technology.[86] To conduct this evaluation, FBI provided access to their TAG-IMAGE database—which includes images of thousands of prisoner tattoos collected from prison inmates and arrestees—to nineteen corporate and academic groups with the goal of developing

85. This Feature adapts its concept of "data relations" from prior work. Nick Couldry and Ulises A. Mejias use the term "data relations" to describe the process of capturing and processing social data, which they argue results in a new social order based on continuous tracking. Nick Couldry & Ulises A. Mejias, *Data Colonialism: Rethinking Big Data's Relation to the Contemporary Subject*, 20 TELEVISION & NEW MEDIA 336, 336 (2018). Data social relations are constituted by both legal and technical systems that influence how data is created, collected, transmitted, and used.

86. Dave Maass, *FBI Wish List: An App that Can Recognize the Meaning of Your Tattoos*, ELEC. FRONTIER FOUND. (July 16, 2018), https://www.eff.org/deeplinks/2018/07/fbi-wants-app-can-recognize-meaning-your-tattoos [https://perma.cc/V48D-DD6V].

image-recognition technologies capable of identifying individuals by their tattoos, as well as identifying tattoos that are markers of various gang affiliations.[87]

Consider a scenario where FBI partners with a company, "TattooView AI," to provide tattoo-recognition products and automated matching—not only identifying a particular individual via their tattoo, but also determining whether their tattoo connotates gang membership more generally.[88] This tool is then used by

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87. TAG-IMAGE is one of several forms of biometric markers included in the Next Generation Identification system used to automate processes of biometric identification capabilities and extend the tracking of biometric markers beyond those included in the FBI's Automated Fingerprint Identification System. The Biometric Center of Excellence (BCOE) is the agency's primary group working to develop biometrics and identity management. BCOE notes that tattoos and other biometric markers have possible uses for law enforcement well beyond purposes of identity verification. Most notably, automated recognition services allow investigators to use a probe or query image to find similar images. "While the value of image-to-image matching technology is obvious from an identification perspective, the benefits of knowing the symbolism and background behind tattoos and graffiti can be equally valuable. From an intelligence standpoint, certain symbols or graffiti may be used to help establish whether an individual is associated with a particular gang, terrorist organization, or extremist group. This may help determine the extent to which the individual or gang poses a threat to law enforcement or the community, and possibly to recognize and link crimes across the country." CJIS Link, *Image-Based Matching Technology Offers Identification and Intelligence Prospects*, FED. BUREAU INVESTIGATION (Dec. 28, 2012), https://www.fbi.gov/services/cjis/cjis-link/image-based-matching-technology-offers-identification-and-intelligence-prospects [https://perma.cc/WU7E-K4WE].

88. In addition to the 2018 National Institute of Standards and Technology-FBI trial, this scenario gains plausibility from three facts.

First, law enforcement already tracks and identifies gang membership on the basis of certain shared tattoos. On gang databases generally, see Stefano Bloch, *Are You in a Gang Database?*, N.Y. TIMES (Feb. 3, 2020), https://www.nytimes.com/2020/02/03/opinion/los-angeles-gang-database.html [https://perma.cc/6CJX-YLR3], which notes that "[s]elf-identifying as a gang member, in addition to tattoos and officers' descriptions of 'gang related' clothing, are used to make a gang distinction." On the Los Angeles Police Department's partnership with Palantir and California Databases, see Caroline Haskins, *Scars, Tattoos, and License Plates: This Is What Palantir and the LAPD Know About You*, BUZZFEED NEWS (Sept. 29, 2020, 3:00 PM ET), https://www.buzzfeednews.com/article/carolinehaskins1/training-documents-palantir-lapd [https://perma.cc/TS5S-SH6V], which asserts that "[w]ith Palantir, police can search for people by name. But . . . they can also search by race, gender, gang membership, tattoos, scars, friends, or family. 'Male, White, Peckerwood Gang, Skull Tattoo.' 'Person, Male, Hispanic, Vineland Boys, Rosary Tattoo'"; and *CalGang®: About the CalGang® Unit*, CAL. OFF. ATT'Y GEN. (2021), https://oag.ca.gov/calgang [https://perma.cc/TM8E-TXHS]. On the United Kingdom's gang database, see *Gangs Violence Matrix*, METRO. POLICE, (2021), https://www.met.police.uk/police-forces/metropolitan-police/areas/about-us/about-the-met/gangs-violence-matrix [https://perma.cc/S2W8-UAQH]. *See also* James A. Densley & David C. Pyrooz, *The Matrix in Context: Taking Stock of Police Gang Databases in London and Beyond*, 20 YOUTH JUST. 11 (2019) (assessing common critiques of gang databases). On the Georgia

law enforcement to identify potential gang members for heightened police observation.

This biometric data flows across several parties—from the initial arrestee to managers of TAG-IMAGE, to third parties such as TattooView AI, to a law enforcement officer who detains a suspected gang member. It also flows across several legal regimes: criminal law, government procurement and trade secrecy law, contract law, and privacy law. Importantly, this flow begins and ends with two human events: first, a person has his tattoo photographed and added to TAG-IMAGE (let's call him Adam); and second, a person with the same tattoo is detained using that image data (let's call him Ben).[89]

Criminal Street Gang Database, see Joshua Sharpe, *Georgia Gang Database Has Law Enforcement Hopeful, Critics Worried*, ATLANTA J.-CONST. (Feb. 5, 2020), https://www.ajc.com/news/crime--law/georgia-gang-database-has-law-enforcement-hopeful-critics-worried/TJelxnLcBflQQeWpTjY6EJ [https://perma.cc/YQ6B-2ATL], which states that "[c]riteria include admitting gang affiliation; having gang tattoos; displaying gang signs personally or in graffiti; wearing clothing, colors, jewelry and/or bandannas believed to be 'gang dress;' possessing or being referenced in gang documents; being seen with gang members; being identified by evidence online, being identified by a reliable source or being arrested on a gang crime charge or being suspected of a gang crime."

Second, law enforcement already partners with private companies to access automated biometric-identity verification and investigation tools. *See* Kashmir Hill, *The Secretive Company That Might End Privacy as We Know It*, N.Y. TIMES (Mar. 18, 2021), https://www.nytimes.com/2020/01/18/technology/clearview-privacy-facial-recognition.html [https://perma.cc/MVD9-Y3SH].

Third, it was recently reported that Palantir already sells capabilities very similar to the hypothetical ones described below to law enforcement. Haskins, *supra*.

89. This Feature will explore several scenarios where the data collected about one person may be used against another person. Through these scenarios, we can examine the social effects of many common and widespread data practices. For example, data about one person may affect another person via the linkage of two datasets, by revealing data about social networks or genetic information that are by definition shared information, or by applying a prediction algorithm trained off of the data of one person and used against another. This relational effect can have a considerable outsized impact. For example, Cambridge Analytica directly collected data from the 270,000 people who downloaded the "thisisyourdigitallife" application. Because Cambridge Analytica was able to receive those people's social-network data (the profiles of their friends and family), they obtained the profile information of about eighty-seven million users (70.6 million in the United States). This information was used to train an ad-targeting program that delivered microtargeted political advertisements to some portion of Facebook's 190 million United States users (as well as users in the United Kingdom and elsewhere), based on their likelihood to respond to a given advertisement. *See* Nicholas Confessore, *Cambridge Analytica and Facebook: The Scandal and the Fallout So Far*, N.Y. TIMES (Apr. 4, 2018), https://www.nytimes.com/2018/04/04/us/politics/cambridge-analytica-scandal-fallout.html [https://perma.cc/UUX4-HGBA]; Hannes Grassegger & Mikael Krogerus, *The Data That Turned the World Upside Down*, VICE (Jan. 28, 2017, 9:15 AM), https://www.vice.com/en/article/mg9vvn/how-our-likes-helped-trump-win [https://perma.cc/9748-

Standard privacy critiques of data flows like this one emphasize not only the significant stakes of this data flow for both Adam and Ben, but also that Adam's data was collected under highly coercive conditions (that is, while Adam was detained in prison) for a purpose (to identify other gang members) with which he may not agree and over which he has no say. Adam's lack of agency at the point of his data's collection is accorded significant moral and legal relevance in critiques of biometric surveillance.[90]

Adam's and Ben's interests are sufficiently aligned such that enhancing Adam's legal rights vis-à-vis TattooView AI would likely also protect Ben's interests. If Adam is granted a robust right to refuse inclusion of his tattoos in TAG-IMAGE, he is likely to exercise that right. Then, his tattoo-image data cannot be used to detain Ben.

But consider an alternative scenario: TattooView AI develops its tattoo recognition algorithm not from the FBI's TAG-IMAGE dataset, but from a dataset it obtained when TattooView AI purchased TattooID. TattooID is a social platform where tattoo enthusiasts can share photos of their tattoos, tag their tattoo artist, and search for designs. Suppose that Adam, a former gang member who regrets his gang involvement, voluntarily shares his tattoo images on TattooID, and tags them as tattoos related to gang affiliation in the hopes that they can help identify other gang members.

In this alternative scenario, individualist conceptions of how this data may harm Adam do not capture the way this data flow affects Ben. Adam was not coerced into sharing this data, but instead did so willingly. Moreover, the purposes to which this data is being applied align with Adam's intent in sharing it and would, in his view, represent a valid outcome. And yet the fact remains that

LBQM]; Issie Lapowsky, *Facebook Exposed 87 Million Users to Cambridge Analytica*, WIRED (Apr. 4, 2018, 5:43 PM), https://www.wired.com/story/facebook-exposed-87-million-users-to-cambridge-analytica [https://perma.cc/V9A4-UW6D]; Sam Meredith, *Facebook-Cambridge Analytica: A Timeline of the Data Hijacking Scandal*, CNBC (Apr. 10, 2018, 9:51 AM EDT), https://www.cnbc.com/2018/04/10/facebook-cambridge-analytica-a-timeline-of-the-data-hijacking-scandal.html [https://perma.cc/3BTL-2QLA]; Owen Bowcott & Alex Hern, *Facebook and Cambridge Analytica Face Class Action Lawsuit*, GUARDIAN (Apr. 10, 2018, 11:45 AM EDT), https://www.theguardian.com/news/2018/apr/10/cambridge-analytica-and-facebook-face-class-action-lawsuit [https://perma.cc/UUN2-FGHE].

90. *See, e.g.*, Complaint at 3, ACLU v. Clearview AI, Inc., No. 9337839 (Cir. Ct. Cook Cnty., May 28, 2020), https://www.aclu.org/legal-document/aclu-v-clearview-ai-complaint [https://perma.cc/3NA9-VJMX] ("The ability to control their biometric identifiers and to move about in public, free from the threat of surreptitious unmasking or surveillance, is essential to Plaintiffs' members, clients, and program participants in Illinois . . . . Clearview has captured more than three billion faceprints from images available online, all without the knowledge—much less the consent—of those pictured.").

Ben faces significant consequences from this data flow. To the extent Ben's interests are of legal, as well as normative relevance, this presents a problem for data-governance law.

B. Mapping Data Social Relations Along Vertical and Horizontal Axes

The relationships that arise among data subjects, data producers, and the third parties impacted by data use can be mapped along two axes.

Along the vertical axis lies the data relation between an individual data subject and an individual data collector (also known as a data processor). The vertical data relation describes the relationship between Adam and TattooView AI, when Adam agrees to the terms of data collection laid out by TattooView AI and shares his data with them. This vertical data relation structures the process whereby data subjects exchange data about themselves for the digital services the data collector provides.

This vertical social relation is expressed both technically via the flow of data from a data subject to a data collector and legally via the contractual terms that structure the terms of exchange between data subject and data collector, as well as the background consumer- and privacy-law regimes that allocate privileges, claims, and duties between the two parties. This vertical data relation is, in some sense, well understood in data-governance law. As will be discussed in greater detail in Part III, proposals for data-governance reform are attentive to how the law governs this vertical relation, how it may structure unequal relations among data subjects and data producers, and how duties and rights between these two parties may be reallocated to address this imbalance.

The horizontal axis describes how data production relates data subjects not to data collectors, but to one another and to others that share relevant population features with the data subject. The relationship between Adam and Ben describes a horizontal data relation.

This horizontal relation is expressed technically through informational infrastructures that make sense of data subjects via group classification and that operationalize classifications to act back on subjects. These technical expressions apprehend (and in apprehending, help to define) the social fact of group identity via shared preferences, social patterns, and behaviors that make people similar to one another. For example, the horizontal data relation between Adam and Ben apprehends a particular social meaning based on their shared tattoo. This horizontal data relation structures a social process whereby a relevant shared feature (that is, a tattoo) is operationalized to make a prediction and define a social meaning (that is, gang membership) and act back on a group member (Ben) according to this grouping.

Horizontal relations are not actually one-to-one relations between data subjects like Adam and those impacted by data flows like Ben. They are population-based relations. For instance, sharing his tattoo image puts Adam in horizontal relation not only with Ben, but also with everyone who has his tattoo and may be acted upon on the basis of this shared feature. The same holds the other way around. Ben is not only in horizontal relation with Adam, but also with everyone who has a relevant population feature in common with him (in this case, his tattoo) and has shared data about this feature with a data collector.[91] To make the discussion below clearer, let's call the group to which Ben belongs "P_{use}," to denote the population of people on which Adam's tattoo image data is used, and refer to the other individuals (the "other Bens") in this population as B_n.[92] We can call the group to which Adam belongs "$P_{collect}$," to denote the population of people from whom tattoo image data is being collected, and refer to other individuals (the "other Adams") in this population as A_n.

These population-level relations give rise to population-level interests along the horizontal relation. For example, we can understand Ben's interest in Adam's data collection as one instance of the more general interest of P_{use} in $P_{collect}$'s data collection. This interest, unlike those along the vertical relation, does not reduce to the individual provenance of the data. Ben's interest in $P_{collect}$'s data sharing is based on the effect that the use of this data will have on him. This use may occur regardless of whether this data was collected from him, from Adam, or from someone else. In this sense, it does not matter *who* the data "came" from, but *what* such data says about Ben, and *how* such meaning is used to act upon Ben. This is the population-level interest Ben (and others like Ben) have in data that apprehends a relevant shared population feature about them. As the example shows, this interest may arise from data Ben shares, data Adam shares, or data someone else shares. Each individual instance of this interest may be weak, but they occur at scale throughout the data-production economy and link individuals to many other individuals via webs of horizontal relation.

One way to understand data governance's unsatisfying response to downstream social effects from data collection, what I call the "sociality problem," is data-governance law's conceptual commitment to individualism (DIM). This commitment focuses the relevant analysis on how data production may harm data subjects and develops legal responses to such harm. While this commitment may result in improvements to the vertical data relation between data subjects like Adam and data collectors like TattooView AI, it does not address the role that horizontal data relations play in producing social value and social risk. This has several significant consequences discussed in greater detail below.

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91. $P_{collect}$ = [Adam, A_1, A_2, A_3 . . .], where A_n = other individuals in $P_{collect}$.

92. P_{use} = [Ben, B_1, B_2, B_3 . . .], where B_n = other individuals in P_{use}.

C. *The Importance of Horizontal Data Relations in the Digital Economy*

While horizontal data relations are minimally relevant to data-governance law, they are central to how data production produces both social value and social risk. Data production for the digital economy is deeply—even fundamentally—relational.

Data flows are quite literally structured, collected, and produced so as to relate people to one another.[93] Data flows are useful when they relate people to one another. Data flows are designed to represent the ways that people are like one another and reveal meaningful things about one another: how we are alike biologically, interpersonally, politically, and economically.[94]

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93. *See* GREEN, *supra* note 7, at 93-103; Rob Kitchin, *Big Data, New Epistemologies and Paradigm Shifts*, BIG DATA & SOC'Y, Apr.-June 2014, at 2 (detailing how Big Data is not simply denoted by volume, but is, among other features, denoted by being relational in nature, which here means "containing common fields that enable the conjoining of different data sets," and flexible, "holding the traits of *extensionality* (can add new fields easily) and *scalability* (can expand in size rapidly)"); Danah Boyd & Kate Crawford, *Critical Questions for Big Data*, 15 INFO., COMMC'N & SOC'Y 662, 670-71 (2012). For more on the utility of data because of its ability to reveal information on others, see, for example, Sebastian Benthall & Jake Goldenfein, *Data Science and the Decline of Liberal Law and Ethics* 7-8 (Oct. 2, 2020) (unpublished manuscript), https://ssrn.com/abstract=3632577 [https://perma.cc/C2Q2-8D9L].

94. Almost all data harvested from an individual person or personal device has the capacity to be relational. Social-media data reveals information (such as preferences and observations) not just about an individual, but also about her social networks. This information can have political as well as social consequences. For example, network data can be used to probabilistically identify support or opposition for a political candidate or position to target political advertising or get-out-the-vote efforts. *See* Robert M. Bond, Christopher J. Fariss, Jason J. Jones, Adam D.I. Kramer, Cameron Marlow, Jaime E. Settle & James H. Fowler, *A 61-Million-Person Experiment in Social Influence and Political Mobilization*, 489 NATURE 295, 295-98 (2012). Network data can be used to predict a credit score and help proactively track and target a disease, suicides, and gun violence. Ben Green, Thibaut Horel & Andrew V. Papachristos, *Modeling Contagion Through Social Networks to Explain and Predict Gunshot Violence in Chicago, 2006 to 2014*, 177 JAMA INTERNAL MED., 326, 326-33 (2017). Genetic data from a consumer genome test reveals information about one's relatives that can help them detect disease early while also placing them in political and legal projects of reconciliation with and reparations for past racial discrimination. *See* ALONDRA NELSON, THE SOCIAL LIFE OF DNA: RACE, REPARATIONS, AND RECONCILIATION AFTER THE GENOME, at xii-xiii (2016). Location data reveals information about one's household. In fact, almost all data harvested from one person that can be used to make a prediction about them or attempt to change their behavior can be utilized, in the form of a behavioral model, to make a prediction or attempt to change the behavior of others. For more on the utility of data because of its ability to reveal information on others, see, for example, Benthall & Goldenfein, *supra* note 93. Institutional-economics literature extols the competitive value of data via tailoring, prediction, personalization, nudging, and marketplace design. *See, e.g.*, MIT TECH. REV. CUSTOM, *supra* note 25, at 2 ("Data is now a form of capital,

Data flows classify and sort people along particular categories of group membership. This process of sorting and classifying is how an individual becomes rendered as a data subject and is how economic value from production is realized. In other words, data about individuals is useful in the digital economy because it helps to define relevant group categories. These categorizations are operationalized to make sense of people on the basis of their classifications and to act back on such insights.

Data about populations is used to develop models to predict and change behavior, to gain intimate consumer or competitor knowledge for market advantage, and to retain greater surplus value.[95] Such activities demonstrate an orientation toward data subjects that recognizes them not as individuals, but as members of groups that are constituted via their shared features and common patterns of behavior.[96] This process recasts people as assemblages of their social relations and group behaviors and apprehends data subjects as patterns of behavior derived from group-based insights. This basic approach is what makes behavioral targeting, prediction tasks, at-scale risk assessment, and modulated feedback systems both possible and profitable.[97]

Data's relationality is central to how data collection produces economic value. This distinguishes the value of Adam's data for the machine-learning (ML) or artificial-intelligence (AI) applications of the contemporary digital economy from the value personal data has for older forms of consumer surveillance (and that inform the law's current approach to data privacy). Prior to the widespread availability of large-dataset computing technology, data about a data subject like Adam may have been valuable because it helped businesses, employers, the government, and insurers know things about Adam. And to some extent, this is still what makes data about Adam valuable.

But what makes data about Adam particularly and distinctly valuable for the contemporary digital economy is its capacity to help companies make predictions or change the behaviors of *others* based on relevant population features they share with Adam—in other words, on the basis of at-scale population-level horizontal

on the same level as financial capital in terms of generating new digital products and services."). This literature aligns with conceptions of human behavior as predictive and probabilistic in cybernetics. *See* JAMES R. BENIGER, THE CONTROL REVOLUTION: TECHNOLOGICAL AND ECONOMIC ORIGINS OF THE INFORMATION SOCIETY (1986). On the concept of data enclosure, see COHEN, *supra* note 3, at 62-63.

95. Salomé Viljoen, Jake Goldenfein & Lee McGuigan, *Design Choices: Mechanism Design and Platform Capitalism*, BIG DATA & SOC'Y, July-Dec. 2021, at 1, 5.

96. *See* JAKE GOLDENFEIN, MONITORING LAWS: PROFILING AND IDENTITY IN THE WORLD STATE 101-02 (2019); FRANK PASQUALE, THE BLACK BOX SOCIETY: THE SECRET ALGORITHMS THAT CONTROL MONEY AND INFORMATION 191-94 (2015); Benthall & Goldenfein, *supra* note 93, at 893.

97. PASQUALE, *supra* note 96.

data relations. It is this relational value of data that drives much of the imperatives to data access, processing, and use. The distinctive feature of ML- and AI-based systems is that they can be used to know things about Adam that Adam does not know, by inferring back to Adam from A_n. And, of greater legal significance (or concern), data from A_n can be used to train models that "know" things about B_n, a population that may not be in any vertical relation with the system's owner. This is the key shift of at-scale data analysis, as compared to prior digital data collection and use approaches that did not have access to the scope and degree of data aggregation, computation capabilities, and inference models that typify digital economic activity in the past decade. It also highlights the importance of horizontal data relations not only for expressing an expanded set of interests in data flows, but also for structuring the incentives of data collectors along vertical data relations with data subjects.

Two implications follow from recognizing the significance of data's relationality in the digital economy. First, conceiving of data's horizontal relationality as incidental to the task of managing data production is wrong. Data's horizontal relationality does result in observable externality effects (from the perspective of the data subject and from that of status quo data governance); however, conceiving of these effects as "external" to the purposes and uses of data that drive entities to transact for it is incorrect.[98] Enacting horizontal relations is not like producing pollution; if polluters could "magic away" pollution they likely would (if only to save themselves some reputational harm). But the same cannot be said for data producers: data's relationality is central to the business of data production and constitutes much of what makes data production economically valuable in the first place.

Second, data's aggregate effects amplify the consequences of this disconnect. In a typical data flow, any one individual's data is essentially meaningless, and the marginal cost of any one individual defecting from collection is very low.[99]

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98. To be clear, the general descriptive claim that data production results in externalities, that is, that data production produces social costs that are not reflected in the individual transaction, is one this Feature agrees with. *See* Solove, *supra* note 40, at 1880-83; Reidenberg et al., *supra* note 74, at 485-86. However, descriptively noting that an externality exists has limited analytic value in illuminating the structural conditions behind, and inegalitarian features of, data production's social effects.

99. *See* Michael Mandel, *The Economic Impact of Data: Why Data Is Not Like Oil*, PROGRESSIVE POL'Y INST. 6 (2017), https://www.progressivepolicy.org/wp-content/uploads/2017/07/PowerofData-Report_2017.pdf [https://perma.cc/Y8NL-DEED] ("[U]nused data, by itself, has uncertain economic value. Its value depends on how it is combined and used with other data."); *see also* POSNER & WEYL, *supra* note 17, at 205-09 (describing the concept of people as "data producers," who, in aggregate, power the digital economy); Alessandro Acquisti & Jens Grossklags, *Privacy and Rationality in Individual Decision Making*, IEEE SEC. & PRIV., Jan./Feb.

Yet in aggregate, data is highly valuable and grows in value the more data can be combined with other kinds of data.[100] Across many different fields of algorithmic development and machine learning—from computer vision to natural language processing to adversarial machine learning—the rule of thumb is that quality and quantity of data in a model's training set are the biggest determinant of overall performance.[101] More data means better models, which results in digital products that make better predictions about both data subjects as well as others who share relevant features with data subjects. Large-scale data collection and aggregation therefore become key competitive advantages in the digital economy.

Treating data's relationality as an accidental byproduct of data creation misdiagnoses a feature as a bug. The combination of relational and aggregate effects from data production drives companies to collect as much data as possible from data subjects. Data subjects are in turn poorly equipped to exert meaningful coercive force back on to data collectors in the face of collectors' strong incentives to obtain such data. However, the issue is not simply a mismatch in the relative coercive power between parties, but rather the wide range of interests that are not represented in these transactions at all, even while the economic benefits of exploiting these interests motivate the data-collection practices of digital firms.

2005, at 24-30 (explaining how incomplete information, bounded rationality, and systemic psychological deviations from rationality affect individual privacy-sensitive behavior); Alessandro Acquisti, Leslie K. John & George Loewenstein, *What Is Privacy Worth?*, 42 J. LEGAL STUD. 249, 249-74 (2013) [hereinafter Acquisti et al., *What Is Privacy Worth?*] (investigating how individuals conduct privacy valuations); Alessandro Acquisti, Curtis Taylor & Liad Wagman, *The Economics of Privacy*, 54 J. ECON. LITERATURE 442, 442-92 (2016) [hereinafter Acquisti et al., *The Economics of Privacy*] (highlighting how consumers' economic analyses of privacy have evolved over time); Imanol Arrieta Ibarra, Leonard Goff, Diego Jiménez-Hernández, Jaron Lanier & E. Glen Weyl, *Should We Treat Data as Labor? Moving Beyond "Free,"* 108 AM. ECON. ASS'N PAPERS & PROC. 38, 38-42 (2017) (arguing for a paradigm where data is not seen as capital, but as labor).

100. For a detailed discussion of how data creates value, see *supra* Section I.A. In the machine-learning community, it is commonly understood that as datasets grow larger and more readily aggregated with other sources of data, they grow more valuable. *See* sources cited *supra* note 94; POSNER & WEYL, *supra* note 17.

101. *See* Pedro Domingos, *A Few Useful Things to Know About Machine Learning*, 55 COMMC'NS ACM 78, 84 (2012) ("More data beats a cleverer algorithm."); *see also* Alon Halevy, Peter Norvig & Fernando Pereira, *The Unreasonable Effectiveness of Data*, 24 IEEE INTELLIGENT SYS. 8, 8-12 (2009) ("A trillion-word corpus . . . captures even very rare aspects of human behavior."); Amandalynne Paullada, Inioluwa Deborah Raji, Emily M. Bender, Emily Denton & Alex Hanna, *Data and Its (Dis)contents: A Survey of Dataset Development and Use in Machine Learning Research* 1 (NeurIPS 2020 Workshop: ML Retrospectives, Surveys & Meta-Analyses, Working Paper, 2020) ("The importance of datasets for machine learning research cannot be overstated."); Chen Sun, Abhinav Shrivastava, Saurabh Singh & Abhinav Gupta, *Revisiting Unreasonable Effectiveness of Data in Deep Learning Era*, *in* PROCEEDINGS OF THE IEEE INTERNATIONAL CONFERENCE ON COMPUTER VISION 843, 843-52 (2017) ("The success of deep learning . . . can be attributed to . . . availability of large-scale labeled data.").

The prevalence of horizontal interests in data thus creates a structural mismatch in vertical relations between data subjects and data collectors. Data subjects possess only a fraction of the interests in a given data flow—and, as described above, many of their interests in information do not reduce to their vertical transaction with a data collector either. Meanwhile, data collectors are highly motivated to collect as much data from as many data subjects as possible in order to realize the considerable benefits that accrue from exploiting the insights of horizontal data relations. Without attending to horizontal relations in data-governance law, the interests they represent and the behaviors they motivate from data collectors cannot be fully accounted. Misdiagnosing these effects as incidental to the task of preventing further privacy erosion risks developing reforms that are not up to the task of disciplining excessive or overly risky data production.

D. The Absence of Horizontal Data Relations in Data-Governance Law

While horizontal data relations are of primary importance in explaining why data collectors develop infrastructures to collect and monetize data flows, they do not feature much, if at all, in how current data-governance law allocates claims, privileges, and duties among actors in the digital economy. Many of the relevant interests in data production that accrue along these population-level relations are unrepresented in data-governance law.

This has both practical and normative implications. First, as a practical matter, the absence of legal interests for horizontal data relations leaves the law out of step with the importance of these relations for the digital economy. As discussed above, this may preclude effective regulation of vertical relations as well. The imperatives to relate individuals along the horizontal axis motivate data collectors and influence the conditions of exchange between them and data subjects. Horizontal relations are therefore relevant to the task of regulating subject-data collector vertical relations.

Second, the absence of horizontal data relations in law may cause data-governance law to miss—or misconceive—how data production results in particular kinds of injustice. Because population-level interests are not represented, data-governance law is not indexing forms of injustice that operate via horizonal relations. This misconception may also lead to regimes of data governance that inadvertently foreclose socially beneficial forms of data production. This second implication is discussed in greater detail below.

The legal marginality of horizonal data relations leaves many consequences of data production unaccounted for in data-governance law. This includes externalities (such as Ben's lack of representation) in how the law accounts the sum of risks and benefits in the data flow from Adam to TattooView AI. But it also